Environmental Science & Technology

pubs.acs.org/est

Viewpoint

Why Water Demand Is No Longer Enough: Rethinking Sustainability in AI Data Centers

Navid B. Saleh*

Cite This: *Environ. Sci. Technol.* 2026, 60, 18383−18386

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



KEYWORDS: *artificial intelligence, hydrologic systems, water demand, heat*

Water has shaped the trajectory of human civilization and technological progress for millennia. As Giulio Boccaletti argues, the relationship between societies and water has often determined the course of civilization.[1] The rapid advancement of artificial intelligence (AI), particularly its scalability, follows a similar trajectory. Although AI is portrayed as digital, and often virtual, infrastructure, its operation depends on physical thermodynamic systems that must dissipate heat generated during computation. Consequently, AI is becoming closely coupled to hydrologic systems. The global expansion of AI both depends on and influences water and the environment, but its needs and impacts remain local. As a result, water demand has emerged as an important consideration in AI infrastructure planning and development.[2,3] However, water demand alone may no longer be sufficient for evaluating the sustainability of AI infrastructure. As cooling systems increasingly rely on recirculation and reuse, the challenge extends beyond water quantity to water compatibility.

The water demand of AI infrastructure originates from one issue: heat. AI data centers operate continuously and at high computational intensity, requiring uninterrupted cooling to maintain stable operation. Recent estimates suggest that U.S. data centers collectively consume approximately 449 million gallons of water per day,[2,3] while future AI expansion may require an additional 697−1451 million gallons per day by 2030.[3] Such demand approaches the average daily municipal water supply of New York City. The underlying challenge, however, is energy rather than water. Water enters the discussion because heat generated during computation must ultimately be dissipated. Consequently, data center operators are pursuing direct-to-chip cooling, immersion cooling, closed-loop systems, and dry-cooling technologies to reduce water consumption. For example, Microsoft has announced next-generation designs that operate without evaporative water loss, while operators such as

Published: June 25, 2026



Downloaded from pubs.acs.org/esthag/article-pdf/60/26/18383/6524479/acs.est.6c08380.pdf by guest on 18 October 2026

ACS Publications

© 2026 American Chemical Society

https://doi.org/10.1021/acs.est.6c08380
Environ. Sci. Technol. 2026, 60, 18383−18386

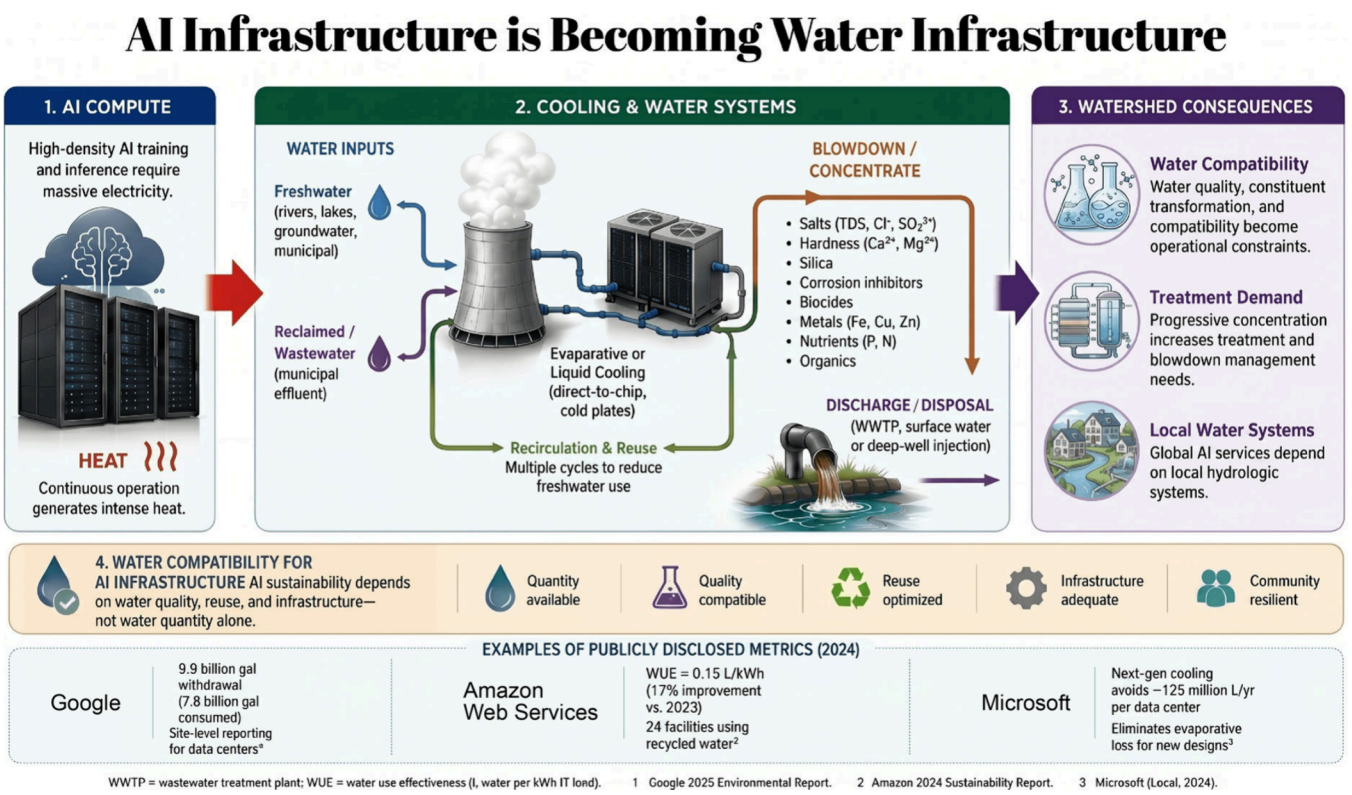


Figure 1. AI infrastructure and associated water management pathways. High-density AI computation requires continuous cooling and increasingly relies on recirculation, reclaimed water use, and blowdown management. Cooling-water recirculation and reuse may result in progressive concentration of dissolved and suspended constituents, with implications for water quality and treatment. Figure 1 was created by graphic artist Asif Hossain.

Novva Data Centers have adopted water-free cooling approaches. These developments may substantially reduce future water demand. However, reducing water demand does not eliminate challenges associated with reuse water, treatment needs, cooling fluids, and long-term system operation. Water demand remains an important metric. It does not, however, capture all sustainability challenges associated with AI cooling systems.

Individual facilities can already consume several million gallons of water per day, with Google reporting annual withdrawals approaching 1.4 billion gallons at a single data center campus in Council Bluffs, Iowa.[4] Although access to these services is global, the infrastructure supporting them is highly localized, requiring in-house cooling and water-management systems (Figure 1). As a result, data center operators are increasingly supplementing freshwater demand with reclaimed wastewater to reduce withdrawals. Circular water systems, either developed from scratch or retrofitted to facilitate water reuse, are emerging as one of the key innovations in AI data center operation and maintenance.[2,4,5] The continuous operation of AI infrastructure is unlikely to change. Thus, cooling water management broadly has two pathways. First, computational density may be reduced or better managed to lower cooling demand; such efforts are already evident through the Advanced Research Projects Agency-Energy (ARPA-E) funded COOLERCHIPS program and related efforts focused on thermal management of high-density computational systems.[6,7] Second, spent water may be treated and recirculated and/or supplemented with wastewater sourced from tertiary effluent, municipal reclaimed water, or treated industrial wastewater.[4]

Evaporative losses are inherent to most circular cooling-water systems and result in progressive concentration of dissolved and suspended constituents. The resulting water chemistry depends strongly on the source water and may include dissolved ions (e.g., trace metals, metalloids, and ligands), dissolved organic carbon (DOC), nutrients, ammonia, hardness, silica, and microbial constituents. Cooling systems commonly operate at cycles of concentration (COC) ranging from approximately 3–10.[8,9] Consequently, dissolved constituents become progressively enriched during recirculation. For example, source waters containing approximately 200 mg L^{-1} total dissolved solids (TDS) may exceed 1000 mg L^{-1} during operation. At these concentrations, scaling, corrosion, silica deposition, and increasingly intensive treatment may become operational constraints even when water quantity is adequate. Progressive enrichment of these constituents ultimately increases treatment demand and complicates blowdown management. Thus, water reuse transforms the challenge from water quantity to compatibility. Water demand asks whether water is available; water compatibility asks whether that water can be used sustainably.

Circular water systems are not unique to AI data centers and are common in power generation, semiconductor manufacturing, and industrial cooling operations. AI data centers differ in several respects. These include their rapid expansion and associated increase in water demand and management needs, their distributed infrastructure making centralized water recirculation and treatment more challenging, and their perception as virtual (e.g., "cloud computing") often masking the physical infrastructure requirements and associated operation and maintenance requirements. Public disclosures of

https://doi.org/10.1021/acs.est.6c08380
Environ. Sci. Technol. 2026, 60, 18383−18386

Downloaded from pubs.acs.org/esthag/article-pdf/60/26/18383/6524479/acs.est.6c08380.pdf by guest on 18 October 2026

major AI leaders demonstrate the increasing water demand and expanding reuse efforts across the industry (Table S1). For example, Amazon Web Services (AWS) has expanded the use of recycled water across multiple facilities. Similarly, semiconductor manufacturers, such as TSMC and Intel, increasingly incorporate reclaimed water use, return flows, and watershed restoration efforts into operation and maintenance (Table S1). Unlike conventional industries where cooling-water demand is expected and geographically constrained, AI services are globally distributed while their water burdens remain local. As a result, AI operators are increasingly becoming participants in local water infrastructure through investments in water reuse, treatment, recycling, and watershed management, thereby extending their role beyond water consumption alone (Table S1).

The transition from freshwater withdrawals to water reuse changes the challenges facing AI data-center cooling systems. This transition presents several engineering challenges. First, insufficient understanding of evaporation and resulting constituent enrichment at each COC leaves water quality control uncertain. Second, progressive enrichment of dissolved and suspended constituents increases the risk of scaling, fouling, corrosion, and operational malfunction during recirculation. Third, depending on the source of the reuse water, opportunities may exist for constituent recovery and partial cost offset.

These challenges suggest that water withdrawals and consumption alone may not adequately capture the sustainability implications of AI cooling systems. Source-water chemistry, reuse fraction, cycles of concentration, blowdown generation, constituent enrichment, and treatment demand may also need attention. They also suggest that treatment design should begin with the chemistry of the circulating water, the cooling technology, and intended reuse objective rather than the adoption of generic reuse strategies. Such treatment may involve advanced separation technologies and/or functional materials for constituent recovery, or simpler treatment units designed to address location-specific needs. As reclaimed wastewater becomes more integrated into AI infrastructure, utilities, municipalities, and data-center operators may need to work more closely to support treatment, reuse, and water-management objectives. Water compatibility therefore becomes a design criterion rather than merely a consequence of reuse.

AI data centers are not only pursuing water reuse and recirculation strategies; some are also developing alternative cooling technologies and fluids that may reduce the need for water cooling altogether. Immersion cooling,[10] dielectric fluids,[11] and direct-to-chip cooling[12] technologies are a few examples of such efforts. These developments are important and may significantly reduce water demand. However, they may also shift environmental concerns from water quantity and quality toward cooling-fluid composition, lifecycle, disposal, and environmental fate. In either case, sustainability extends beyond resource consumption alone and requires consideration of the materials and fluids used to sustain AI operation.

Current discussions surrounding AI sustainability have largely focused on water demand and freshwater withdrawals. This viewpoint argues that such considerations, while important, may no longer be sufficient as cooling systems increasingly adopt recirculation, reuse, and alternative cooling technologies. The challenge facing AI is no longer solely one of water demand. As cooling systems evolve, the challenge shifts toward water quality, constituent transformation, compatibility, and the treatment systems required to manage them. Sustainability assessments may therefore need to expand beyond resource consumption alone to include water chemistry, constituent enrichment, treatment demand, and the broader infrastructure needed to support AI operation. Boccaletti's hydrologic governance perspective during this period of technological growth may therefore be understood not merely as a digital transition, but as a hydrologic transition. Water has shaped the trajectory of human civilization for millennia. The future trajectory of AI may similarly depend not only on computational capability, but also on how effectively society understands and manages the hydrologic systems that sustain it. Sustainability assessments that focus only on water demand risk overlooking the compatibility constraints that may ultimately govern AI cooling systems.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.6c08380.

Publicly reported water-use, water-reuse, freshwater withdrawal, water-consumption, and cooling-related metrics for major AI infrastructure operators and semiconductor manufacturers, including Google, AWS, Microsoft, Meta, Equinix, TSMC, and Intel, with associated references (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Navid B. Saleh − *Fariborz Maseeh Department of Civil, Architectural and Environmental Engineering, The University of Texas at Austin, Austin, Texas 78712, United States;* orcid.org/0000-0001-6092-5783; Phone: (512) 471-9175; Email: navid.saleh@utexas.edu

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.est.6c08380

Notes

The author declares no competing financial interest.

Biography



Navid B. Saleh is the Jewel McAlister Smith Endowed Professor in the Fariborz Maseeh Department of Civil, Architectural and Environmental Engineering at The University of Texas at Austin. He received his Ph.D. from Carnegie Mellon University and completed postdoctoral training at Yale University. His research advances empathic engineering approaches to environmental challenges by integrating community knowledge, human-centered design, and technological innovation. His work spans water quality, climate-change adaptation, environmental

Downloaded from pubs.acs.org/esthag/article-pdf/60/26/18383/6524479/acs.est.6c08380.pdf by guest on 18 October 2026

https://doi.org/10.1021/acs.est.6c08380
Environ. Sci. Technol. 2026, 60, 18383−18386

nanotechnology, and sustainable infrastructure systems, with ongoing collaborations involving Indigenous and local communities in the United States, Mexico, and Bangladesh. Professor Saleh has authored more than 100 peer-reviewed publications and currently serves as an Editor of the *Journal of Hazardous Materials*. He is a Past President of the Sustainable Nanotechnology Organization and serves on the Editorial Boards of *Environmental Science & Technology* and *Environmental Science: Nano*. His recent work explores how emerging technologies interact with environmental systems and how sustainability considerations can be incorporated into technology development and deployment.

■ ACKNOWLEDGMENTS

I sincere acknowledge graphic designer Asif Hossain for digitally rendering Figure [1].

■ REFERENCES

(1) Boccaletti, G. *Water: A Biography*; Vintage, 2022.
(2) Mytton, D. Data Centre Water Consumption. *npj Clean Water* **2021**, *4* (1), 11.
(3) Han, Y.; Li, P.; Wierman, A.; Ren, S. Small Bottle, Big Pipe: Quantifying and Addressing the Impact of Data Centers on Public Water Systems. *arXiv* **2026**, DOI: 10.48550/arXiv.2603.02705.
(4) Environmental Report 2025. Google, 2025 (https://sustainability.google/reports/google-2025-environmental-report/) (accessed 2025-05-17).
(5) Kseibati, R. Cooling Innovation and Circularity: Addressing Water Stress in the Age of AI-Driven Data Centers. Master's Thesis, Massachusetts Institute of Technology, Cambridge, MA, 2025 (https://dspace.mit.edu/entities/publication/75a37ce0-3f8e-4524-990c-126e6a96050d).
(6) Booth, H. UTA team exploring new technology to cool data centers. The University of Texas at Arlington (https://www.uta.edu/news/news-releases/2023/05/12/doe-agonafer-computer-cooling) (accessed 2026-05-18).
(7) Advanced Research Projects Agency-Energy (ARPA-E). COOLERCHIPS. ARPA-E (https://arpa-e.energy.gov/programs-and-initiatives/view-all-programs/coolerchips).
(8) Cooling Towers. In *016 ASHRAE Handbook—HVAC Systems and Equipment*; ASHRAE: Atlanta, 2016.
(9) Soliman, M.; Eljack, F.; Kazi, M.-K.; Almomani, F.; Ahmed, E.; El Jack, Z. Treatment Technologies for Cooling Water Blowdown: A Critical Review. *Sustainability* **2022**, *14* (1), 376.
(10) Kim, J.; Choi, H.; Lee, S.; Lee, H. Computational Study of Single-Phase Immersion Cooling for High-Energy Density Server Rack for Data Centers. *Appl. Therm. Eng.* **2025**, *264*, 125476.
(11) Chhetri, A.; Kashyap, D.; Mali, A.; Agarwal, C.; Ponraj, C.; Gobinath, N. Numerical Simulation of the Single-Phase Immersion Cooling Process Using a Dielectric Fluid in a Data Server. *Second Int. Conf. Sustain. Energy Solut. Better Tomorrow SESBT 2021* **2022**, *51*, 1532−1538.
(12) Ahmadi, V. E.; Barestrand, H. A.; Summers, J.; Yilmaz, C. Performance Evaluation of Two-Phase Direct-to-Chip Liquid Cooling Combined with Air Cooling for Data Centers. *Case Stud. Therm. Eng.* **2026**, *78*, 107650.

Downloaded from pubs.acs.org/esthag/article-pdf/60/26/18383/6524479/acs.est.6c08380.pdf by guest on 18 October 2026

https://doi.org/10.1021/acs.est.6c08380
Environ. Sci. Technol. 2026, 60, 18383−18386